

Soft Spaces – Phase 3.2

v32.15: Robustness and Generalization of the Degeneracy-Closed Interference Law

Technical checkpoint note – 10 September 2026

Purpose

Version v32.15 tests whether the basis-invariant hotspot mechanism established in v32.14 is specific to the frozen 11-term Hamiltonian or survives moderate changes in model complexity. The scientific question is therefore

$$\boxed{\text{Does the degeneracy-closed interference/coherence signal survive model variation?}} \quad (1)$$

The robustness scan varies only the Hamiltonian term count,

$$N_{\text{terms}} \in \{7, 9, 11\}, \quad (2)$$

while retaining the 8-qubit setting, the 12-seed ensemble, complete degeneracy-closed eigenspaces, and the dephasing and transverse perturbation families. The 11-term case remains the frozen reference.

Basis-invariant mechanism under test

For a target eigenspace P and a complete degeneracy-closed complementary eigenspace G , define

$$T_G(E) = g_\eta(E - E_G) P V P_G V P, \quad (3)$$

with

$$g_\eta(x) = \frac{x}{x^2 + \eta^2}, \quad \Sigma_E = \sum_G T_G(E). \quad (4)$$

The normalized interference parameter is

$$\boxed{A_E^{(G)} = \frac{2 \sum_{G < K} \text{Re Tr}(T_G^\dagger T_K)}{\left(\sum_G \|T_G\|_F \right)^2}} \quad (5)$$

and the associated coherence factor is

$$\boxed{C_E^{(G)} = \frac{\|\Sigma_E\|_F}{\sum_G \|T_G\|_F}}. \quad (6)$$

Both quantities are invariant under basis rotations within degenerate eigenspaces.

Results across Hamiltonian term counts

The primary diagnostic is the seedwise Spearman rank correlation with frozen hotspot prominence. Table 1 summarizes the median seedwise correlation and the number of positive seeds.

Family	Metric	7 terms	9 terms	11 terms
Dephasing	$A_E^{(G)}$	+0.3452 (10/12)	+0.4722 (10/12)	+0.2605 (10/12)
Dephasing	$C_E^{(G)}$	+0.5119 (11/12)	+0.6471 (11/12)	+0.2500 (12/12)
Transverse	$A_E^{(G)}$	+0.8048 (11/12)	+0.4353 (10/12)	+0.2843 (10/12)
Transverse	$C_E^{(G)}$	+0.7333 (11/12)	+0.3415 (8/12)	+0.5194 (12/12)

Here “(k/12)” denotes the number of seeds with positive correlation.

For $A_E^{(G)}$, the positive direction is preserved for *every* tested term count and both perturbation families. The one-sided sign-test values are significant for all six $A_E^{(G)}$ tests: $p = 0.01929$ for the 10/12 cases and $p = 0.00317$ for the 11/12 transverse 7-term case.

The pooled rank correlations are also positive throughout. For example, transverse $A_E^{(G)}$ gives

$$\rho_{\text{pooled}} = \begin{cases} +0.6615, & 7 \text{ terms,} \\ +0.3544, & 9 \text{ terms,} \\ +0.2915, & 11 \text{ terms.} \end{cases} \quad (7)$$

Weakest robustness point

The weakest case is transverse $C_E^{(G)}$ at 9 terms:

$$\rho_{\text{median}} = +0.3415, \quad 8/12 \text{ positive seeds,} \quad p_{\text{sign}} = 0.19385. \quad (8)$$

This case does not pass the seed-significance criterion by itself. Importantly, the interference parameter $A_E^{(G)}$ remains positive and significant in the same 9-term transverse ensemble:

$$\rho_{\text{median}} = +0.4353, \quad 10/12 \text{ positive seeds,} \quad p_{\text{sign}} = 0.01929. \quad (9)$$

The stress case therefore weakens the coherence-factor evidence but does not remove the underlying interference signal.

Interpretation

The v32.15 result changes the status of the Phase 3.2 mechanism. After v32.14, the mechanism was basis-invariant but still demonstrated only in the frozen 11-term ensemble. v32.15 shows that the same positive interference direction survives moderate changes in Hamiltonian complexity.

The strongest compact statement supported by the present scan is

$$\boxed{\Pi(E) \text{ tends to increase with constructive, degeneracy-closed } P \rightarrow Q \rightarrow P \text{ interference}} \quad (10)$$

within the tested 7-, 9-, and 11-term model class.

This is not evidence of universality over arbitrary Hamiltonians. It is, however, evidence that the effect is not confined to one finely tuned 11-term realization.

Phase 3.2 checkpoint after v32.15

The current evidence can now be organized in three layers:

1. **Global structure:** the $\pm E$ organization is explained by the algebraic symmetry $\Gamma H \Gamma^\dagger = -H$.

2. **Local mechanism:** hotspot strength is associated with constructive interference among complete virtual Q_G channels, described by $A_E^{(G)}$ and $C_E^{(G)}$.
3. **Robustness:** the positive mechanism survives changes from 11 to 9 and 7 Hamiltonian terms.

v32.15 robustness conclusion: the degeneracy-closed interference mechanism is not confined to the frozen 11-term Hamiltonian and remains structurally visible across the tested model variations.

Recommended next step

The next step should no longer be another broad feature search. The most useful continuation is a stricter generalization boundary test: enlarge the seed ensemble and vary the regularization scale η over a narrow pre-specified range, while keeping $A_E^{(G)}$ as the primary basis-invariant endpoint. This would determine how far the robustness result can be extended before Phase 3.2 is closed as a theory section.